Angles of elevation and depression

1 α or $\angle ACB$

2 σ or $\angle CBD$

3 $x = 37.23^\circ$

4 $h = 19\text{ m}$

5 $d = 285\text{ m}$

6 $x = 53.13$

7

a $AC = 8663\text{ m}$ **b** $BC = 6975\text{ m}$ **c** 1688 m

8

a $x = 11.86\text{ m}$ **b** $y = 16.24\text{ m}$ **c** $BC = 4.4\text{ m}$

9 $a = 40.24^\circ$

10 $\theta = 52.57^\circ$

11 $\theta = 30.00^\circ$

12 $y = 43.41^\circ$

13 $y = 41.09^\circ$

14 $d = 4\text{ m}$

15 $d = 1.14\text{ m}$

16

a $x = 32.78^\circ$ **b** $y = 6.22^\circ$ **c** $z = 10.27^\circ$ **d** 28 cm

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$d = 229\ \text{m}$

18 $h = 9\ \text{m}$

19 $h = 17\ \text{m}$

20 $d = 10.86\ \text{m}$

21 $\theta = 45.97^\circ$

22 $d = 814.32$

23

a $x = 259.79\ \text{m}$ **b** $238.79\ \text{m}$

24

a $h = 20.80\ \text{m}$ **b** $22.2\ \text{m}$

25

a $221.12\ \text{m}$ **b** $110.73\ \text{m}$ **c** $110.4\ \text{m}$

26 $x = 28^\circ$

27 $\theta = 74.33^\circ$

28 $19.56\ \text{m}$

29

a $x = 15.98\ \text{m}$ **b** $y = 62.1\ \text{m}$ **c** $\theta = 20^\circ$

Additional questions

30 $h = 19\ \text{m}$

31 $\theta = 43.79^\circ$

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32 $y = 54.90^\circ$

33 $\theta = 52.52^\circ$

34 $\theta = 41.81^\circ$

35 $\theta = 16.26^\circ$

36 $x = 182 \text{ m}$

37 $\theta = 23.9^\circ$

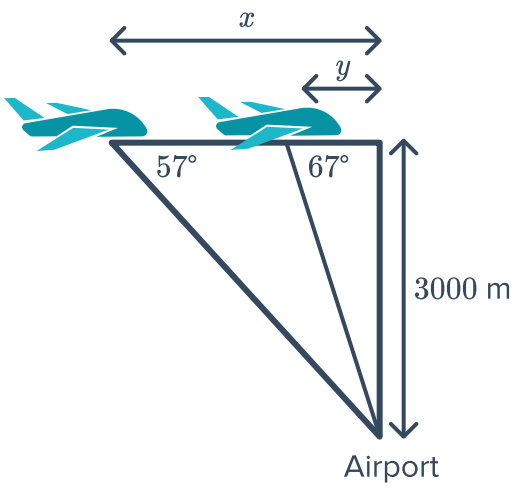
38 $d = 814 \text{ ft}$

39 $x = 38^\circ$

40 $\theta = 74.27^\circ$

41

a



b $x = 1948.2 \text{ m}$

c $y = 1273.4 \text{ m}$

42

a 225 ft

b 232.5 m

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